Human Activity Classification Based on Cognitive Doppler Radar to Optimize Carrier Frequency and Sampling Rate Using Reinforcement Learning

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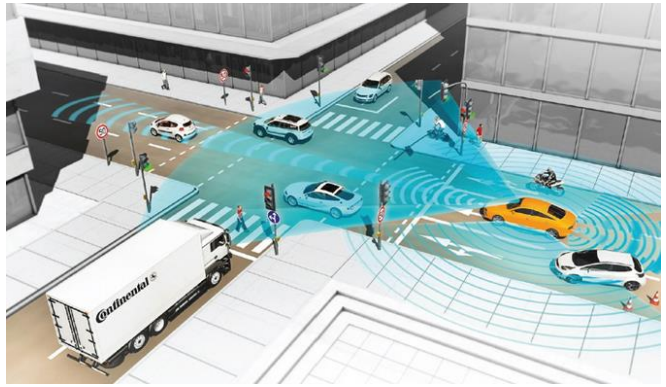
Application of Human Monitoring Using EM

Security and Surveillance

Disaster search and rescue operation, physical activity, law enforcement, and border patrol



Autonomous Vehicle

1. Background & Motivation

Classification Techniques for Spectrogram

- Prior Works

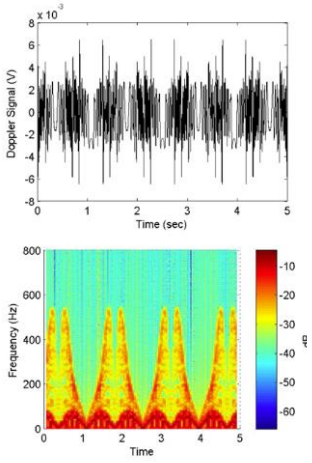


Fig. 3 *(Top left) simulated micro-Doppler signal, (top right) experi (bottom right) STFT of experimental signal for human swinging arm*

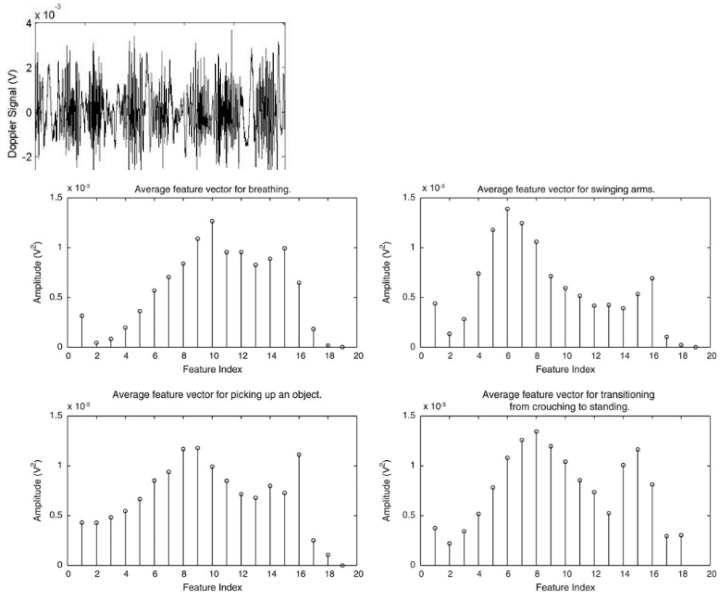


Fig. 8 *Averaged feature vectors for a human (top left) breathing, (top right) swinging arms, (bottom left) picking up an object and (bottom right) standing up from a crouching position for the W-band radar*

Specific Vectors Utilization [1]

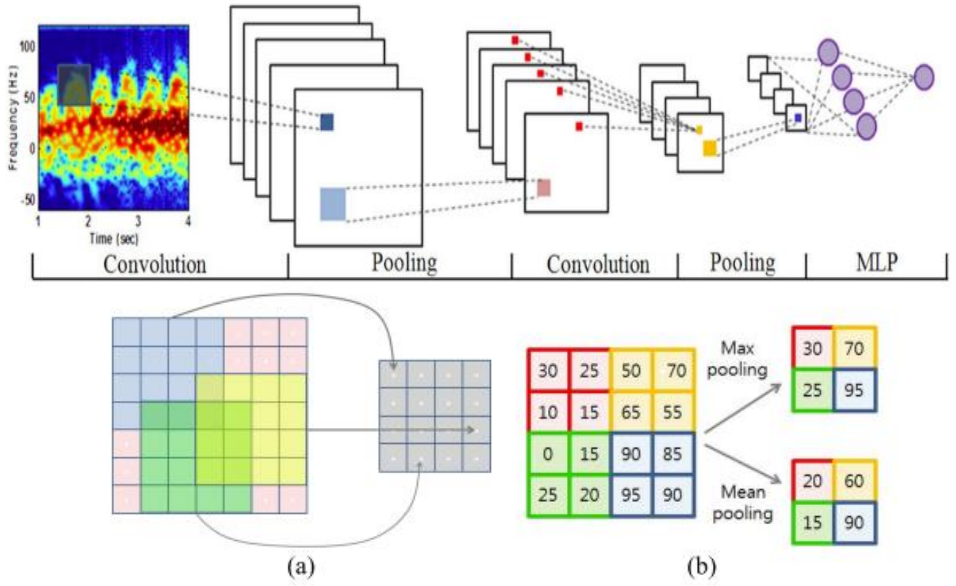


Fig. 2. (a) Process of applying a 4 × 4 convolution filter to the input data (in pink) to generate the output (in gray). (b) Examples of 2 × 2 pooling (max or mean pooling) that reduces the data dimension by half.

Deep Learning(DCNN, RNN, GAN) Approaches [2]

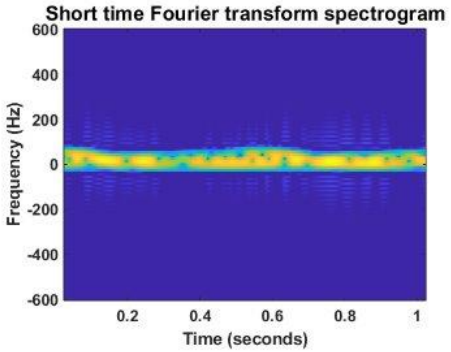
Focused on fixed radar parameter conditions

[1] S. Björklund, H. Petersson, A. Nezirovic, M. B. Guldogan, and F. Gustafsson, “Millimeter-wave radar micro-Doppler signatures of human motion,” in Proc. 12th Int. Radar Symp. (IRS), Leipzig, Germany, Sep. 2011, pp. 167–174.

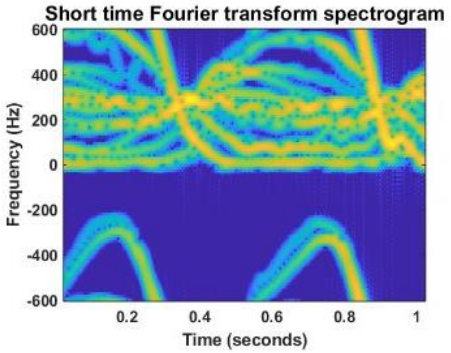
[2] Y. Kim and T. Moon, “Human detection and activity classification based on micro-Doppler using deep convolutional neural networks,” IEEE Geosci. Remote Sens. Lett., vol. 13, no. 1, pp. 8–12, Jan. 2016.

Spectrogram Characteristics

- **Carrier Frequency**
  - **Walk Motion Spectrogram**

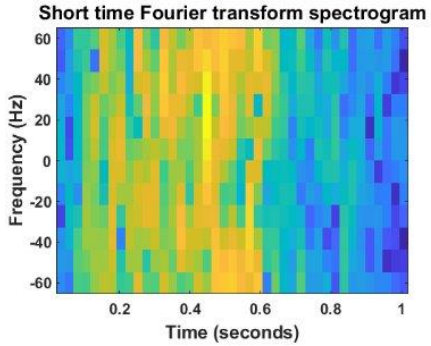


$f_C = 1.5GHz, f_S = 1200Hz$

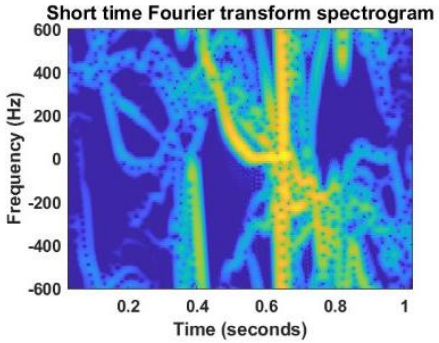


$f_C = 30GHz, f_S = 1200Hz$

- **Sampling Rate**
  - **Run Motion Spectrogram**



$f_C = 15GHz, f_S = 200Hz$



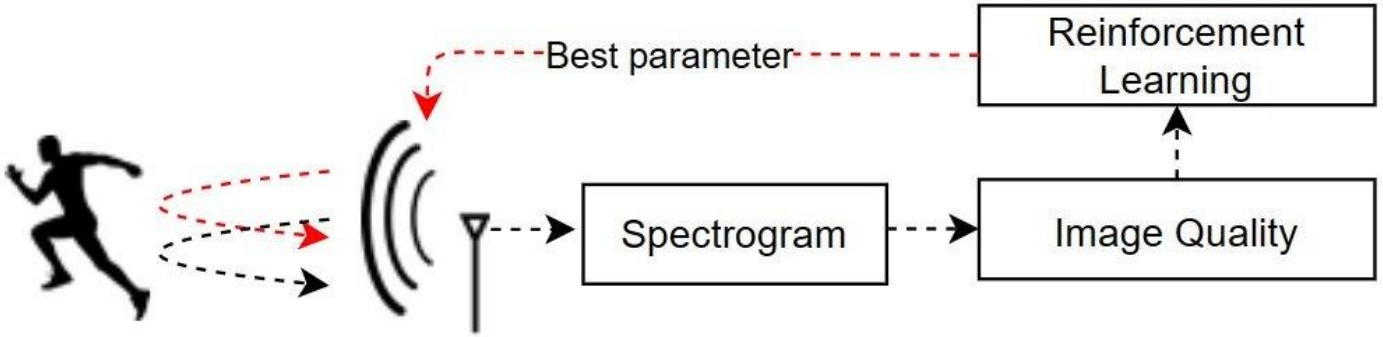
$f_C = 15GHz, f_S = 1200Hz$

Why Cognitive Radar?

Spectrogram Images Affects the Performance of DCNN

+

Optimal Parameters for spectrogram differs based on human motions



<u>Intelligent System for Adaptive Radar Parameter Adjustment based on Prior Knowledge from Previous Observations and Databases</u>

Roadmap

Phase 1 : Analyze the relationship between radar parameters and DCNN performance

Phase 2 : Analyze the relationship between image quality and DCNN performance

Phase 3 : RL model based on previous analysis

Roadmap

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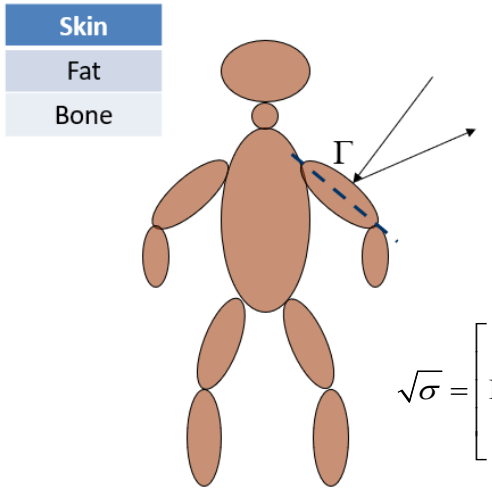
Doppler Simulator using Computer Animation

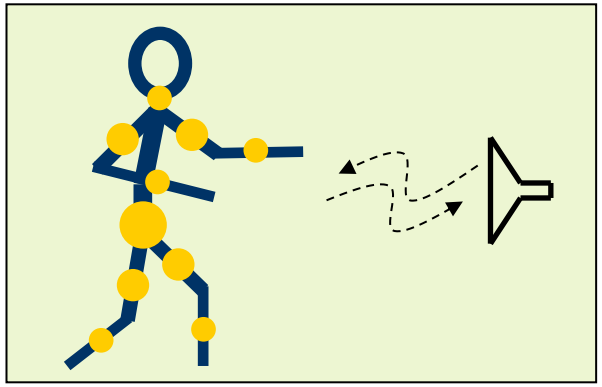
Poser

BVH file

BVH file describes all the positions of body parts with respect to time.

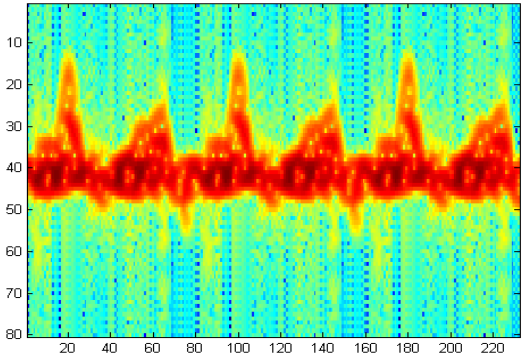


$$\sqrt{\sigma} = \left[\Gamma \cdot \frac{\frac{1}{4}\pi \,\mathrm{Re}^4\, He^2}{\mathrm{Re}^2 \sin(\theta) + \frac{1}{4} He^2 \cos(\theta)} \right] \cdot e^{-j\frac{2\pi}{\lambda}2r}$$



Freq. and radar position are adjustable.

Independent of Hardware

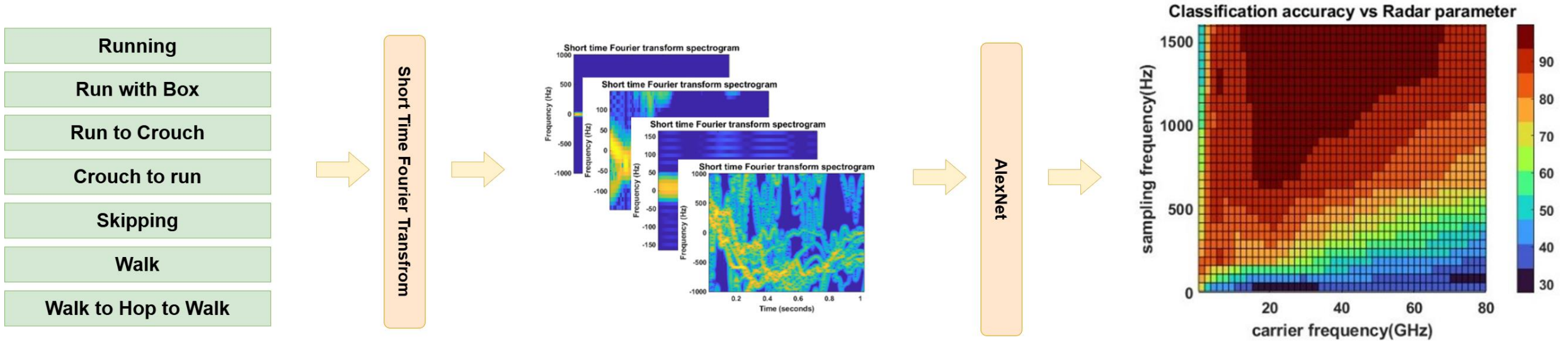


Each bone has its own Radar Cross Section(RCS)

Doppler Simulator[1]

[1] Ram, C. Craig, Y. Kim and H. Ling, "Simulation and analysis of human micro-Dopplers in through-wall environments," IEEE Transactions on Geoscience and Remote Sensing, vol. 4, pp. 2015-2023, April 2010

Classification Accuracy & Radar Parameters



Radar parameters vs Classificiation accuracy Heat map generation using Alexnet

- Alexnet Properties
  - # for each motion : 50
  - Input size = [227 227 3]
  - Train data : Validation data = 0.76 : 0.24
- Parameter Conditions
  - Carrier frequency : logscale 312.5MHz ~ 80GHz
  - Sampling rate : logscale 10Hz ~ 2560Hz
  - Radar is placed collinearly with the motion

Specific ranges carrier frequency and sampling rate that ensure high DCNN performance exists

Roadmap

Phase 1 : Analyze the relationship between radar parameters and DCNN performance

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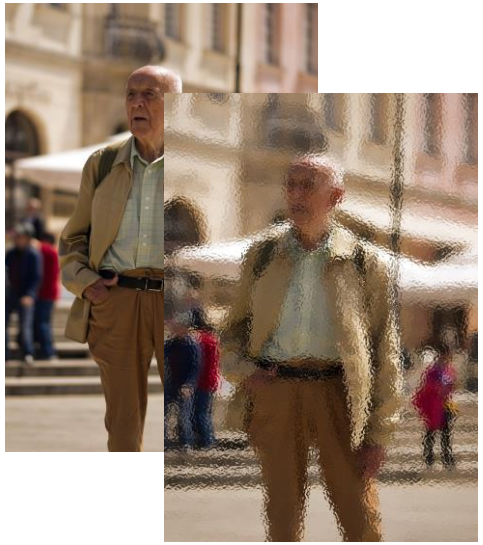
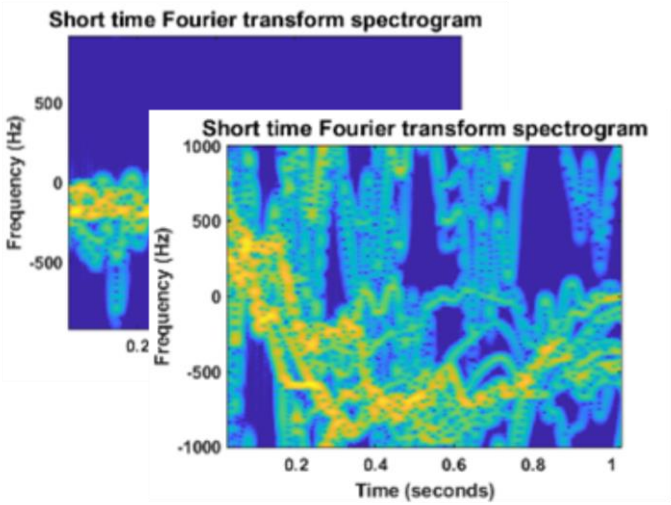
Phase 3 : RL model based on previous analysis

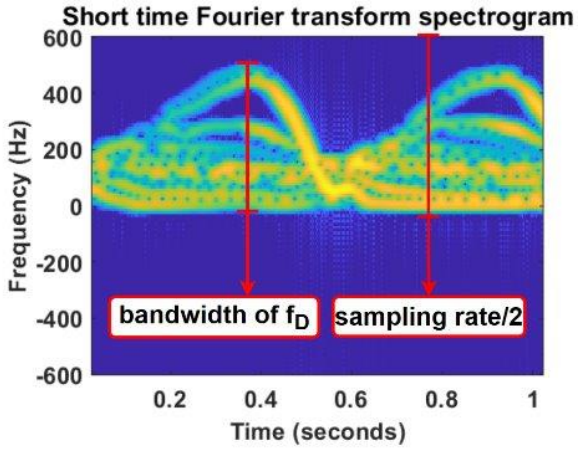
Spectrogram Image Quality





- **Folding Rate**

$$\text{Folding rate} = \frac{2fD}{Sampling\ rate} \times 100$$



- **Definition Rate**

$$\text{Definition rate} = \text{Sampling rate.}$$

Classification Accuracy & Image Quality

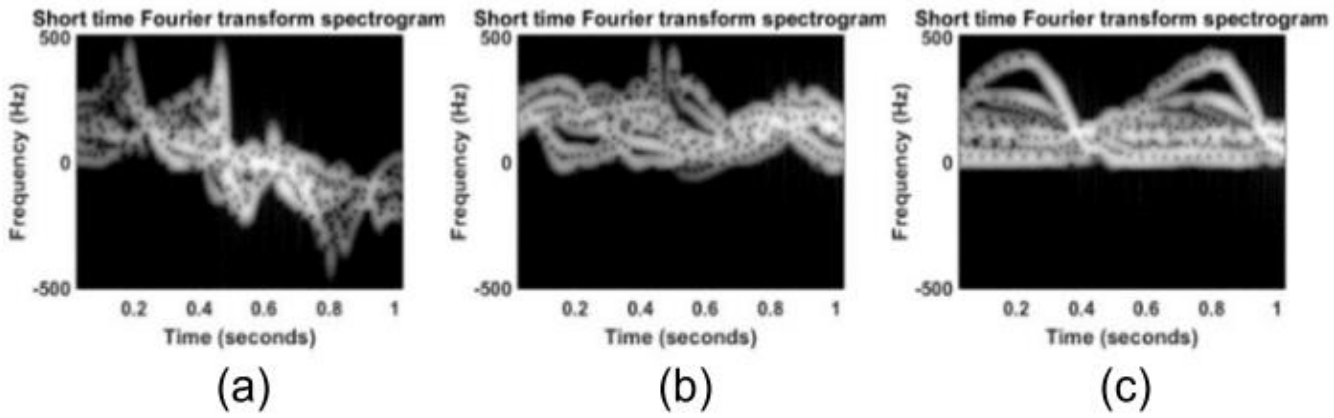
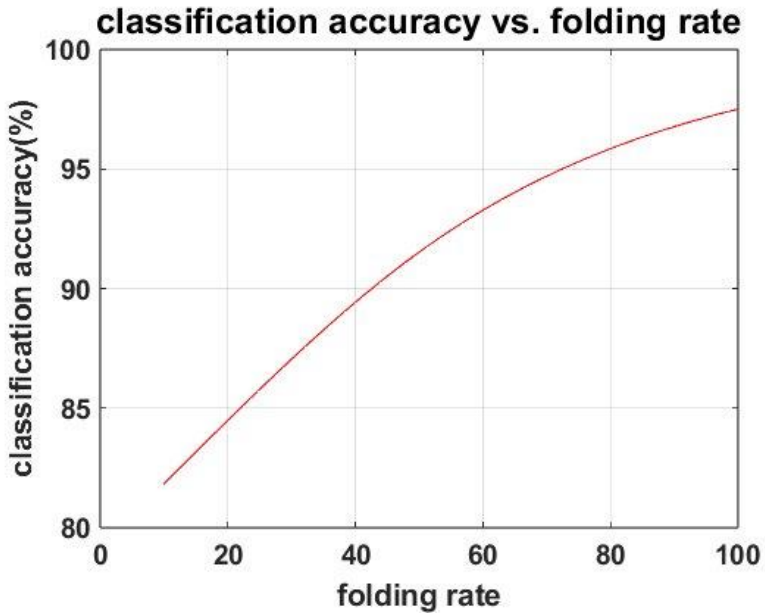


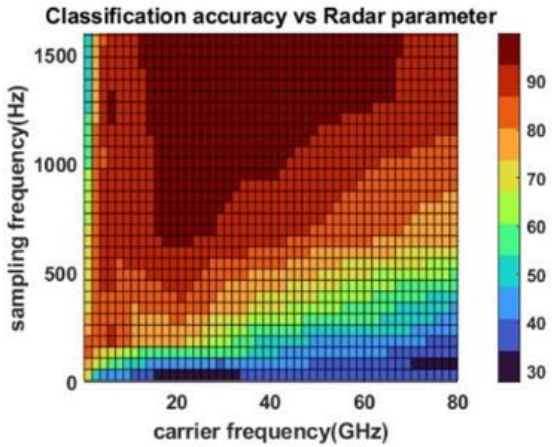
Fig. 8. Human motion spectrogram whose folding rate is 90. (a) Run. (b) Skipping. (c) Walk.

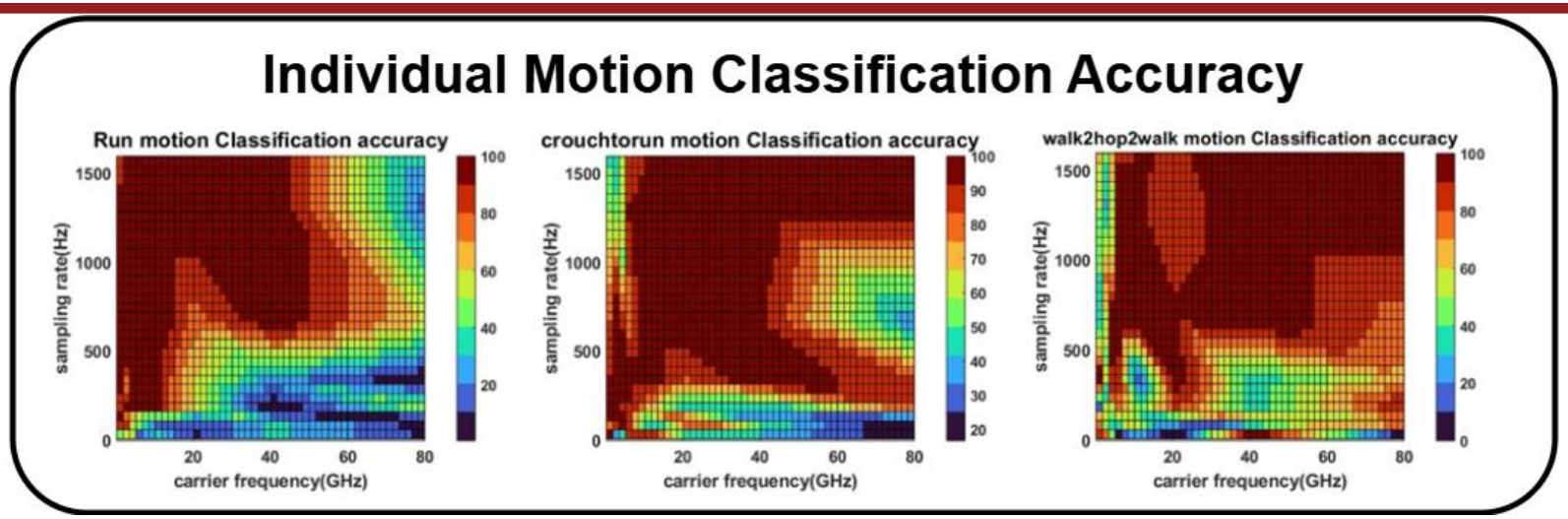


- Alexnet Properties
  - # for each motion : 50
  - Input size = [227 227 3]
  - Train data : Validation data = 0.76 : 0.24
- Parameter Conditions
  - Folding rate : linscale 10 ~ 100
  - Sampling rate : 1kHz
  - Radar is placed collinearly with the motion

Near 100 Folding Rate Correlate with Improved DCNN performance

Classification Accuracy & Image Quality





- Higher classification performance is observed when the carrier frequency results in above 90
- There needs to have a certain level of definition rate depending on the motions

TABLE I
RADAR PARAMETER (f_c) WHEN HUMAN MOTION SPECTROGRAM FOLDING RATE IS 90 AND f_s IS 1 kHz

Human motion	Carrier frequency(GHz)
Run	4.2
Run with box	3.6
Run to crouch	6
Crouch to run	4.8
Skipping	11
Walk	13.5
Walk2hop2walk	6

High Image Quality is closely associated with better classification accuracy

Roadmap

Phase 1 : Analyze the relationship between radar parameters and DCNN performance

Phase 2 : Analyze the relationship between image quality and DCNN performance

Phase 3 : RL model based on previous analysis

Reinforcement Learning for Cognitive Radar

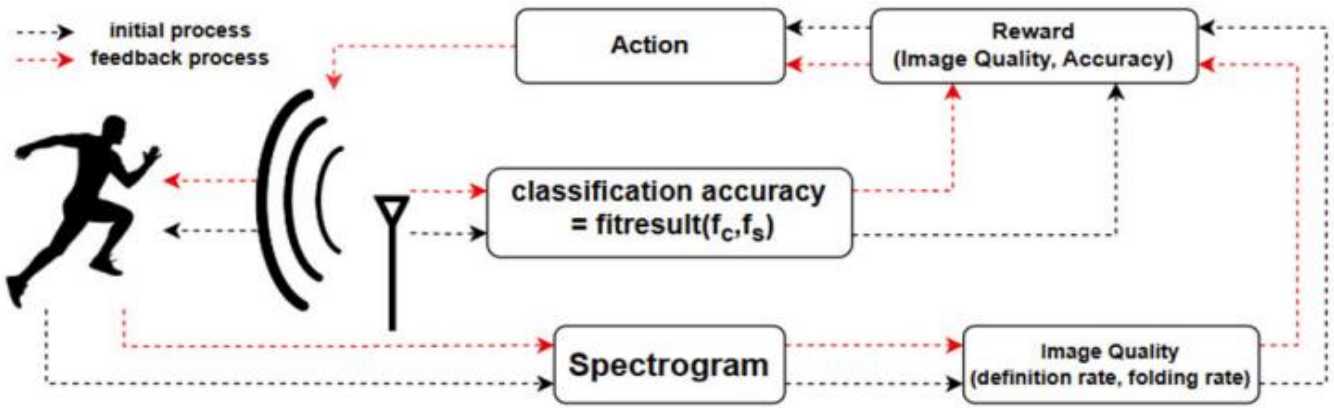


Fig. 12. Structure of RL for training the cognitive radar.

- Environment Settings
  - Max carrier frequency : 80GHz
  - Max sampling rate : 2kHz
- Action Settings
  - 8 discrete actions

TABLE II
EIGHT POSSIBLE COMBINATIONS IN RL ALGORITHM

Radar parameter	Increase	Decrease
Carrier frequency(GHz)	0.5 or 2	0.5 or 2
Sampling rate (Hz)	50 or 500	50 or 500

Reinforcement Learning for Cognitive Radar

- Algorithm : Q-Learning
- Two Q-functions
  - Q(carrier frequency, sampling rate)
  - Q(folding rate, definition rate)
- Reward Function
  - R(folding rate, definition rate, accuracy, step)

$$Q_{freqsamp}(S_t, A_t) \leftarrow (1-\alpha) \cdot Q_{freqsamp}(S_t, A_t) + \alpha[R_t + \gamma \cdot Q_{freqsamp}(S_{t+1}, A_t)]$$

$$Q_{foldblur}(S_t, A_t) \leftarrow (1-\alpha) \cdot Q_{foldblur}(S_t, A_t) + \alpha[R_t + \gamma \cdot Q_{foldblur}(S_{t+1}, A_t)]$$

<Bellman Equation>

$$A \leftarrow \begin{cases} argmax_a \left\{ Q_{freq_{samp}}(S, A) + Q_{blur_{fold}}(S, A) \right\} & \epsilon \geq \varepsilon \\ random\ action & \epsilon < \varepsilon \end{cases}$$

<Epsilon-Greedy Policy>

- Reward Function Summary

	Desired Range	Target	Penalty	Reward
Folding Rate	90-100	100	Difference from 100, scaled by the difference	Closer to 100 = Higher reward
Definition Rate	500-1000	500	Difference from 500, scaled by observed definition rate	Exceeds 500 = Higher reward
Accuracy	N/A	Improve	Decreases in accuracy scaled by 5x step size	Improvement scaled by 5x step size

4. Results

RL verification

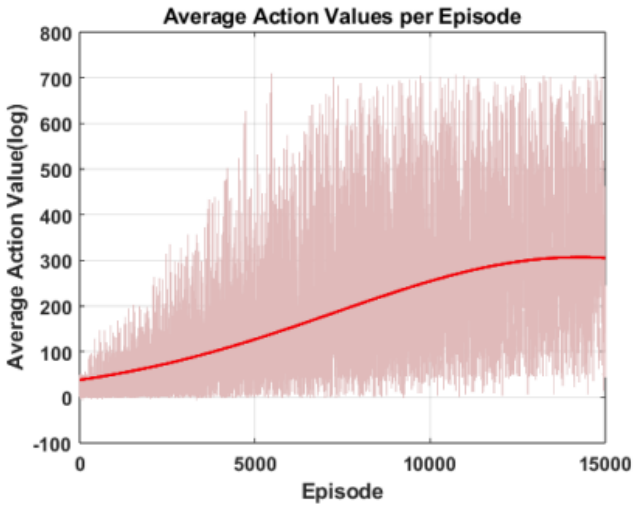
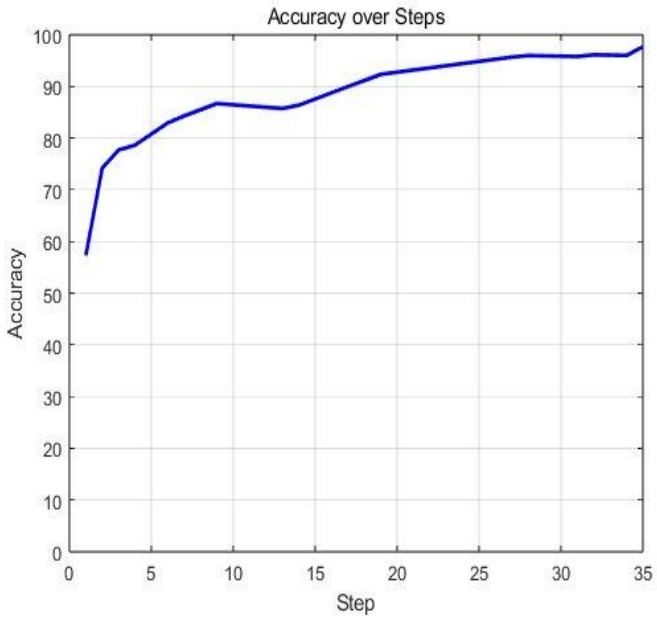
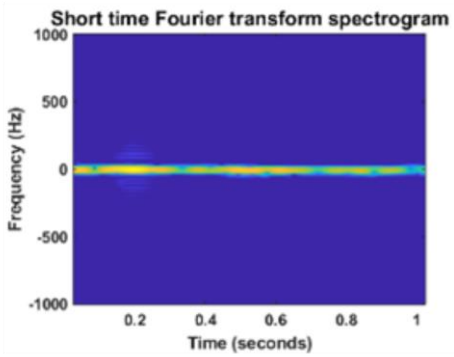
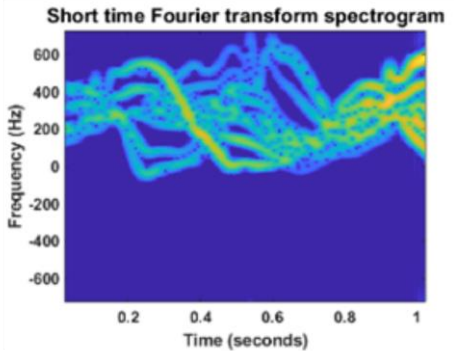


Fig. 14. Learning curve of RL.

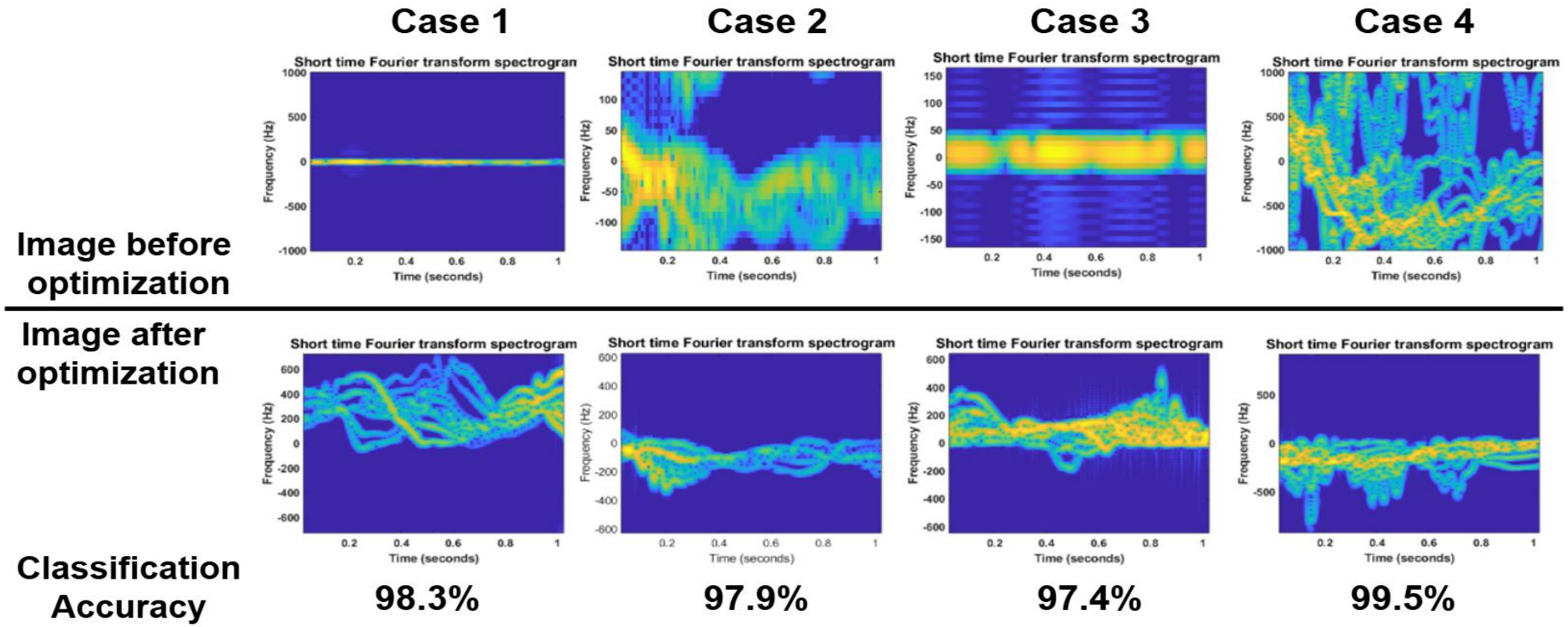






- RL verification
  - Learning curve analysis
  - Worse case environment settings
    - Before : fc = 1GHz, fs = 2000Hz, Folding Rate = 2, Definition Rate = 2000
    - After : fc = 17GHz, fs = 1450Hz, Folding Rate = 92, Definition Rate = 1450
    - Accuracy : 59% → 98.3%

RL verification



RL effectively optimizes radar parameters to improve DCNN performance

Conclusion & Future Tasks

- The application of reinforcement learning to cognitive radar systems effectively learns and optimizes radar parameters, leading to improved accuracy in classifying micro-Doppler signals.
- The adjustments of radar parameters to optimize spectrogram quality contributes to enhancing classification performance

- Future tasks
  - Radar Performance Enhancement Across Diverse Domains Using Complex Motion Patterns
  - Application of different RL algorithm(Actor-Critic, etc.) to minimize the iteration

Thank You